MIL-C-14747 (AR)
AMENDMENT 4
17 November 80
Superseding
Amendment 3
31 July 1977

MILITARY SPECIFICATION

CARTRIDGE, 105MM: APERS-T, M494
FOR 105MM GUN M68
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-14747 (AR) dated 11 July 1968.

PAGE 1

Title, paragraph 1.1 and paragraph 2.1, drawings, Army: Delete the nomenclature "XM494E3" and "XM603E1" wherever it appears and substitute "M494" and "M603", respectively.

PAGE 2

** Add to the list of Drawings the following:
"9278696 - Inspection of M86 Detonators for presence of pettman cement when installed in M494 fuze adaptor."

Paragraph 3.4.1: Delete the paragraph in its entirety and substitute the following new paragraph:

"3.4.1 Functioning. - The cartridges shall function satisfactorily as evidenced by first and second projectile function. The first projectile function shall occur prior to 50 feet from the muzzle when the fuze is set MA and no earlier than 230 meters of later than 380 meters when the fuze is set at the 400 meter marking."

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PAGE 3

Paragraph 3.4.2, line 2: Delete the words "at ambient temperature" and substitute "at 70 degrees Fahrenheit".

Paragraph 3.5.4: Delete in its entirety and substitute the following:

"3.5.4 Adhesive on threads. - Threads shall be clean and free of contaminants prior to the application of the specified quantity of adhesive, (see 4.3.4)."

PAGE 6

** Paragraph 4.2.2.3, Major Defect 102: Delete in its entirety and substitute the following:
"Pettman cement on any end of detonators...Gage 03002"

Major Defect 103: Delete in its entirety.

PAGE 7

Paragraph 4.2.2.5, add Major Defect 103, 100 percent inspection:

"103. Distance from surface "X" and bay assembly, bay 5, minimum.....Gage 05004"

PAGE 8

Paragraph 4.2.2.7, Major Defect 101: Delete in its entirety and substitute the following:

"101. Adhesive on threads inadequate (3 Places) (see 3.5.4 & 4.3.4).....Visual/Gage 07001"

PAGE 9

Paragraph 4.2.2.10, Major Defect 103: Delete in its entirety and substitute the following:

"103. Primer after staked fails to withstand disassembly torque (see 4.3.1).....Torque Test 10010"

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Paragraph 4.2.2.10, add Major Defect 104:

"104. Crimp not present and end of cartridge case not rolled over second rotating band for 360°.....Visual 10011"

PAGE 11

Paragraph 4.2.3.2, first sentence: Delete the words "After the adhesive has set" and substitute "After the cement has cured."

PAGE 12

Paragraphs 4.2.3.7, 4.2.3.7.1, 4.2.3.7.1.1 and 4.2.3.7.2: Delete these paragraphs in their entirety and substitute the following:

"4.2.3.7 Proving Ground

4.2.3.7.1 MA Function - Major Defect. - Ten Sample cartridges shall be randomly selected from each lot for this test. The sampling plan and procedures for this test shall be in accordance with Standard MIL-STD-414, Section B, Part I, Form 2, Table B-3, Code letter F an an AQL of 1.5 percent. In addition, if a total of two or more sample cartridges malfunction, [(i.e., duds, unsatisfactory functions (i.e. cartridges don't exhibit first and second projectile functioning) or outliers (see 6.6)], the lot shall be rejected.

4.2.3.7.2 400 Meter Function - Major Defect.

4.2.3.7.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with all of the requirements in 3.4, 25 sample cartridges shall be randomly selected from each lot for this test. The sampling plan and procedures for this test shall be in accordance with Standard MIL-STD-414, Section B, Part II, Table B-3, Code Letter I and an AQL of 1.5 percent. In addition, if a total of three or more sample cartridges malfunction, [i.e., duds, unsatisfactory functions (i.e. cartridges don't exhibit first and second projectile functioning) or outliers (see 6.6)], the lot shall be rejected.

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4.2.3.7.2.2 After three consecutive lots have complied with 4.2.3.7.2.1, ten sample cartridges shall be randomly selected from each lot for this test. The sampling plan and procedures for this test shall be in accordance with Standard MIL-STD-414, Section B, Part II, Table B-4, Code Letter I and an AQL of 1.5 percent. In addition, if a total of two or more sample cartridges malfunction, [i.e. duds, unsatisfactory functions (i.e. cartridges don't exhibit first and second projectile functioning) or outliers (see 6.6)], the lot shall be rejected. In the event that three successive lots fail to meet all of the criteria of 4.2.3.7.2.2, the sampling provisions of 4.2.3.7.2.1 shall again apply. When three successive lots have again successfully passed all of the requirements specified therein, the provisions of 4.2.3.7.2.2 shall be reinstated.

PAGE 13

Paragraphs 4.2.3.7.3, 4.2.3.7.4, 4.2.3.7.5 and 4.2.3.7.6: Delete these paragraphs in their entirety and substitute the following:

4.2.3.7.3 Muzzle Velocity - Major Defect. - All rounds tested for 400 meter function shall be measured for muzzle velocity.

4.2.3.7.3.1 Those samples cartridges fired in accordance with the increased sample size of 4.2.3.7.2.1 shall be subjected to the sampling plan and procedure specified in Standard MIL-STD-414, Section B, Part II, Table B-3, Code Letter I and an AQL of 1.5 percent. In addition, if the standard deviation of the sample cartridge muzzle velocities multiplied by the factor 0.75 exceeds the applicable requirement, the lot shall be rejected.

4.2.3.7.3.2 Those sample cartridges fired in accordance with the reduced sample size of 4.2.3.7.2.2 shall be subjected to the sampling plan and procedure specified in Standard MIL-STD-414, Section B, Part II, Table B-4, Code Letter I and an AQL of 1.5 percent. In addition, if the standard deviation of the sample cartridge muzzle velocities multiplied by the factor 0.65 exceeds the applicable requirement, the lot shall be rejected.

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4.2.3.7.4 Tracer Function and Dye Marker Effectiveness - Major Defect. - All rounds fired shall be observed for tracer function and dye marker effect. If three or more of the sample cartridges tested in accordance with the increased sample size of 35 fail to comply with either the tracer requirement of 3.4.3 or the dye marker requirement of 3.4.4, the lot shall be rejected. If two or more of the sample cartridges tested in accordance with the reduced sample size of 20 fail to comply with either the tracer requirement of 3.4.3 or the dye marker requirement of 3.4.4, the lot shall be rejected.

4.2.3.7.5 Security - Critical Defect. - If any cartridge fails to comply with the requirement of 3.4.5, the lot shall be rejected.

PAGE 14

Paragraph 4.3.4: Delete in its entirety and substitute the following:

"4.3.4 Adhesive on Threads (see dwg. 9229964). - The adequacy of the manufacturing application method shall be quantitatively assessed on one sample at the start of each production shift, as well as each time there is a change in the method or process that is likely to affect uniformity of performance as determined by the Government. If the sample fails to meet the drawing requirement, production shall be stopped and the application method corrected. When it has been determined that the quantity applied by the application method in use meets the drawing requrement, visual observation shall be used to assure that this amount is applied to each of the threaded joints, (see 4.2.2.7, Major Defect 101), (See also 6.3)."

Paragraph 4.3.5.1: Delete in its entirety and substitute the following new paragraphs:

"4.3.5.1 MA Function. - The sample cartridges fired in this phase shall be verified for proper MA fuze setting. Adequate photgraphic coverage shall be provided to determine compliance with the applicable requirements. The distance from the muzzle to the point of second function shall be recorded

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for information. The temperatures of the sample cartridges at the time of firing shall be recorded for information.

4.3.5. 2 400 Meter Function. - The sample cartridges fired in this phase shall be conditioned to 70 plus or minus 2-1/2 degrees Fahrenheit for a minimum of 24 hours prior to testing and fired with the fuze set for 400 meters. Radar or other appropriate instrumentation shall be provided to determine compliance with the applicable requirements. Distance to second function shall be recorded for information.

4.3.5.3 Tracer Function and Dye Marker Effect. - The tracer and dye marker shall be readily observable without the use of instruments."

PAGE 17

After paragraph 6.5, add the following new paragraphs:

"6.6 Outlier Criteria. - An outlier test shall be conducted to determine if any of the first functioning distances are outliers. Any value determined to be an outlier shall be excluded from MIL-STD-414 calculations for lot acceptability. The 'Dixon Criteria and Procedure' for the treatment of outliers as specified in AMCPM 706-113 (Experimental Statistics), paragraph 17-3.2.1, with an upper percentile of .95, shall be utilized.

6.7 Proving Ground Test Summary

TEST	SAMPLES	SIZE	REQUIREMENTS
MA Function	* 10	10	a. First Function (3.4.1) b. Tracer Function (3.4.3) c. Dye Marker Effect (3.4.4) d. Security (3.4.5) e. Second Function Distance for Information (4.3.5.1) f. Temperature for Information (4.3.5.1)

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400 Meter	* 25	10	a. First Function (3.4.1) b. Muzzle Velocity (3.4.2) c. Tracer Function (3.4.3) d. Dye Marker Effect (3.4.4) e. Security (3.4.5) f. Second Function Distance for Information (4.3.5.2)
	* 35	20	

* Applicable for first three lots or tightened inspection.

** The margins of this Amendment are marked with an asterisk to indicate where changes (aditions, modifications, corrections, deletions) from previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue."

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